

Network Performance – Delay

Module 2.2

READING

- Reading: 1.4

Learning Goals

Delay

- List and define the types of delay and how they contribute to over-all delay
- Calculate the end-to-end delay in a network.
- Perform bottleneck analysis on a path
- Compute traffic intensity and relate traffic intensity to queuing delays
- Calculate link utilization
- Use the formula of Average Delay = $S/(1-U)$ where U is the network utilization and S is the average service time for a single packet (this formula only applies to randomly arriving packets)

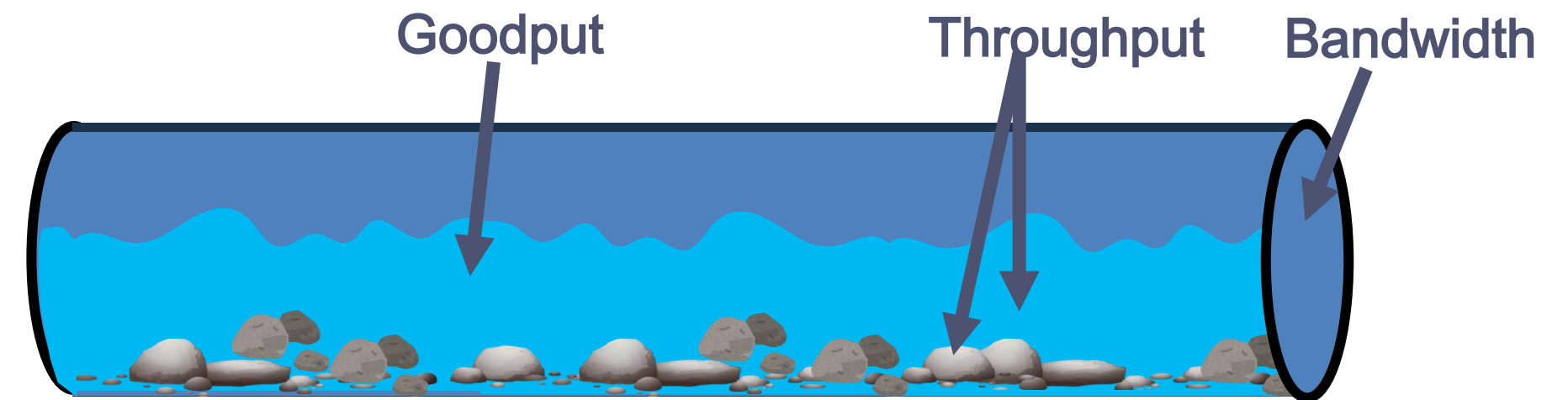
Recap

- Bandwidth
- Throughput
- Goodput

Rates
measured in
bits per
second

- Latency
- RTT (Round Trip Time)
- Jitter

Times
measured in
seconds



Clicker Question

If lightning happens 1km from me, it takes about 3 seconds for me to hear the thunder. What time is that?

- A. Latency
- B. RTT
- C. Bandwidth
- D. Throughput
- E. Goodput

Clicker Answer

If lightning happens 1km from me, it takes about 3 seconds for me to hear the thunder. What time is that?

- A. Latency
- B. RTT
- C. Bandwidth
- D. Throughput
- E. Goodput

Clicker Question

Suppose you have a 5 GB movie that you want to download on a 100Mbps link. How long will it take (rounded to the nearest second)? Assume 75Mbps goodput.

- A. 573 seconds
- B. 533 seconds
- C. 66 seconds
- D. 50 seconds
- E. None of the above

Clicker Answer

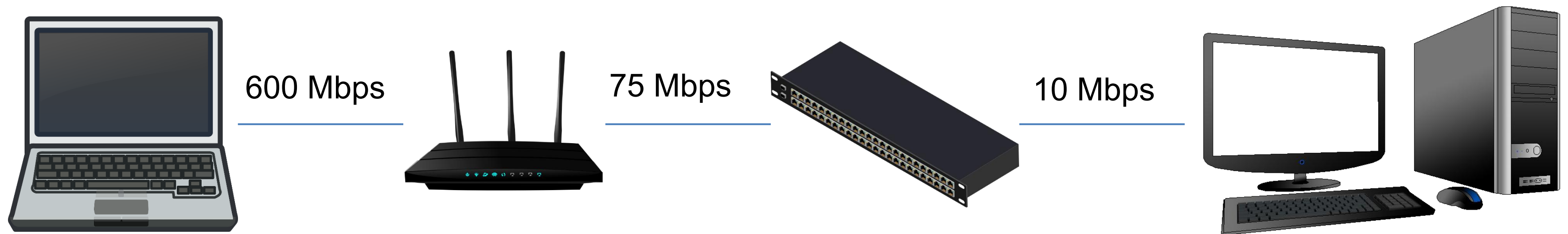
Suppose you have a **5 GB movie** that you want to download on a 100Mbps link. How long will it take (rounded to the nearest second)? Assume **75Mbps goodput**.

$$(5 * 2^{30} * 8) / (75 * 10^6) = 572.66 \text{ seconds}$$

- A. 573 seconds**
- B. 533 seconds
- C. 66 seconds
- D. 50 seconds
- E. None of the above

Bottlenecks

- What is the maximum throughput (possible bandwidth) between two nodes connected by a network?
 - Can traffic flow at maximum bandwidth in all links?

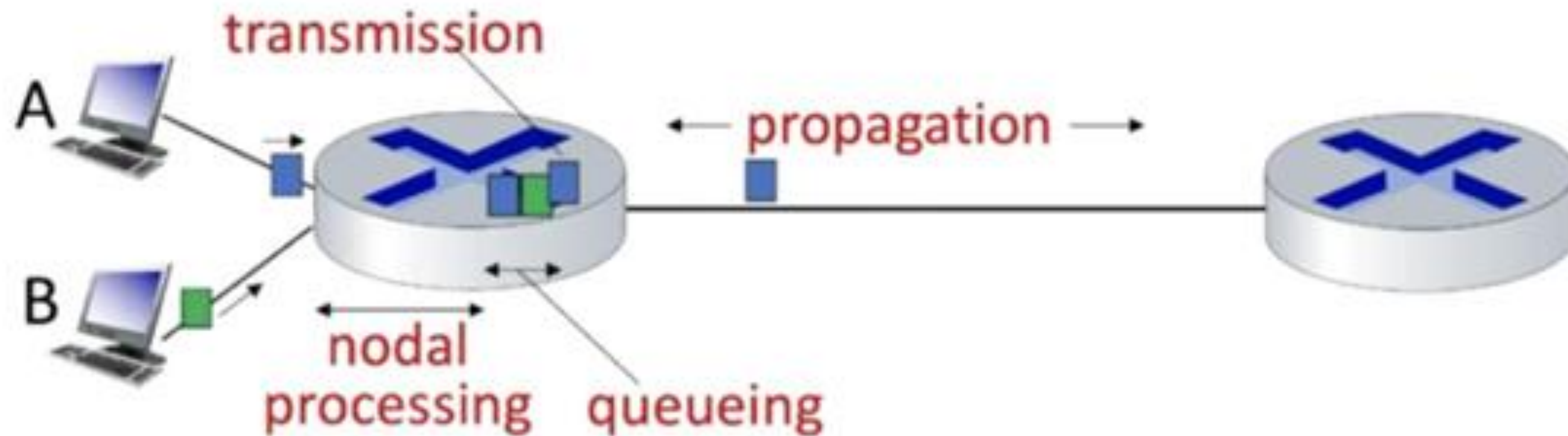


Types of Delay

- *Processing delay*: examine packet to decide where to direct it
- *Queueing delay*: waiting time to get access to the link
- *Transmission delay*: time to actually write the packet onto the medium
- *Propagation delay*: time spent to move each bit from source to destination on the transmission medium
- *End-to-end delay*: sum of all sources of delay

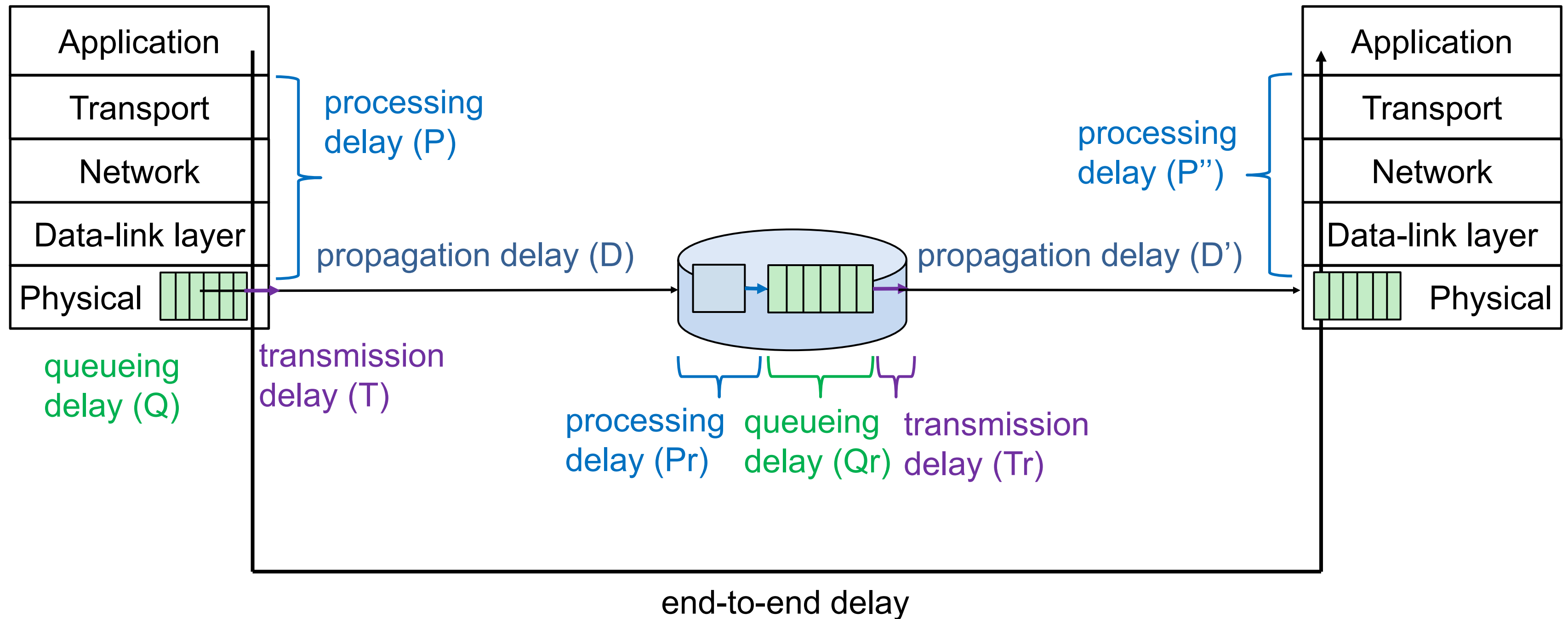


Types of Delay



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

Types of Delay



Classification of Delay: fixed vs variable

- | | | |
|-----------------------------|--------------------|--------------------|
| • <i>Processing delay</i> | | fixed or variable? |
| • <i>Queueing delay</i> | | fixed or variable? |
| • <i>Transmission delay</i> | } <i>same link</i> | fixed or variable? |
| • <i>Propagation delay</i> | | fixed or variable? |
| • <i>End-to-end delay</i> | | fixed or variable? |

Classification of Delay: fixed vs variable

• <i>Processing delay</i>		fixed or variable?	fixed (essentially)
• <i>Queueing delay</i>		fixed or variable?	variable
• <i>Transmission delay</i>	} <i>same link</i>	fixed or variable?	fixed (bit) variable (packet)
• <i>Propagation delay</i>		fixed or variable?	fixed (metre) variable (location)
• <i>End-to-end delay</i>		fixed or variable?	variable

Suppose ...

You are sending the same sized packets to the same destination over and over again.

- *Processing delay.* ?
- *Queueing delay.* ?
- *Transmission delay.* ?
- *Propagation delay.* ?
- *End-to-end delay.* ?

Suppose ...

You are sending the same sized packets to the same destination over and over again.

- *Processing delay.* fixed
- *Queueing delay.* variable
- *Transmission delay.* fixed
- *Propagation delay.* fixed
- *End-to-end delay.* variable

Delay Calculations – Propagation

You are designing a satellite network. The satellites are 750km from the surface of the earth. The speed of light is 3×10^8 metres per second. Packets are 1250 bytes. **What is the two-way propagation delay?**



Delay Calculations – Propagation

- You are designing a satellite network. The satellites are 750km from the surface of the earth. The speed of light is 3×10^8 metres per second. Packets are 1250 Bytes. What is the two-way propagation delay?
- Two-way propagation delay = $2 * \text{one-way propagation delay}$
- One-way propagation delay = distance / speed
- $= (750 * 10^3) / (3 \times 10^8)$
- $= 0.0025 \text{ seconds} * 1000 = 2.5 \text{ ms}$
- Two-way propagation delay = $2 * 2.5 = 5 \text{ ms}$

Delay Calculations – Transmission

You are designing a satellite network. The satellites are 750km from the surface of the earth. The speed of light is 3×10^8 metres per second. Packets are 1250 bytes. The network has a transfer rate (bandwidth) of 100Mbps. **What is the transmission delay (one-way)?**



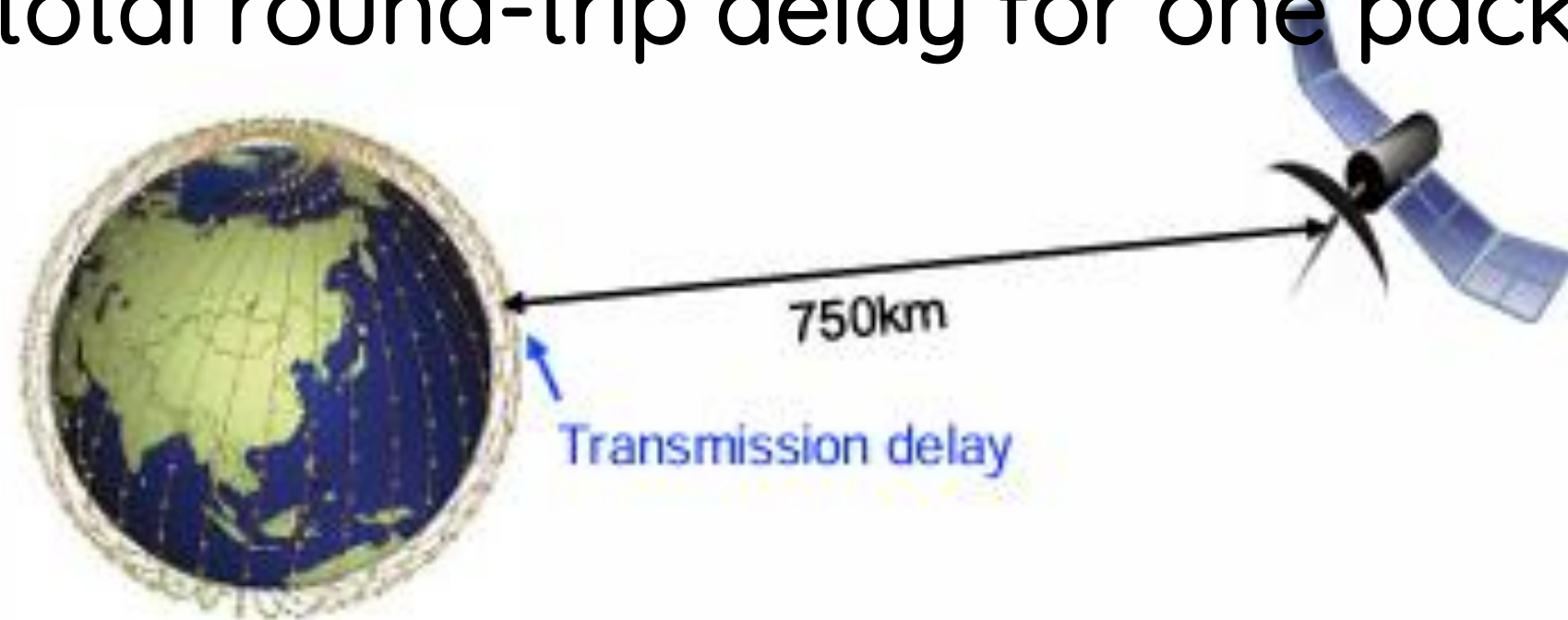
Delay Calculations – Transmission

- You are designing a satellite network. The satellites are 750km from the surface of the earth. The speed of light is 3×10^8 metres per second. Packets are 1250 Bytes. The network has a transfer rate (bandwidth) of 100Mbps. What is the transmission delay (one-way)?

- **Transmission delay = packet size / Transmission rate**
- **= $(1250 \times 8) / (100 \times 10^6)$**
- **= 0.0001 seconds**
- **= 0.1 ms**

Delay Calculations – End-to-end

You are designing a satellite network. The satellites are 750km from the surface of the earth. The speed of light is 3×10^8 metres per second. Packets are 1250 bytes. The network has a transfer rate (bandwidth) of 100Mbps. Assume the transmission delay for the ACK is 0ms, and that the processing and queueing delays are 0. **What is the total round-trip delay for one packet?**



Delay Calculations – End-to-end

- You are designing a satellite network. The satellites are 750km from the surface of the earth. The speed of light is 3×10^8 metres per second. Packets are 1250 Bytes. The network has a transfer rate (bandwidth) of 100Mbps. Assume the transmission delay for the ACK is 0ms, and that the processing and queueing delays are 0. What is the total round-trip delay for one packet?
- $RTT = 2 * \text{propagationDelay} + \text{transmissionDelay}_{\text{packet}}$
- $= 2 * 2.5 + 0.1$
- $= 5.1 \text{ ms}$

More Calculations – Throughput

You are designing a satellite network. The satellites are 750km from the surface of the earth. The speed of light is 3×10^8 metres per second. Packets are 1250 bytes. The network has a transfer rate (bandwidth) of 100Mbps. Assume that the processing and queueing delays are 0. Assume also that you can transmit packets back-to-back.

What is the throughput?

More Calculations – Throughput

- You are designing a satellite network. The satellites are 750km from the surface of the earth. The speed of light is 3×10^8 metres per second. Packets are 1250 Bytes. The network has a transfer rate (bandwidth) of 100Mbps. Assume that the processing and queueing delays are 0. Assume also that you can transmit packets back-to-back. What is the throughput?
- **Throughput = 100 Mbps**

More Calculations – Throughput

You are designing a satellite network. The satellites are 750km from the surface of the earth. The speed of light is 3×10^8 metres per second. Packets are 1250 bytes. The network has a transfer rate (bandwidth) of 100Mbps. Assume that the processing and queueing delays are 0. Assume also that you don't transmit the $n+1^{\text{th}}$ packet until you get a (very short) ACK for the n^{th} packet (assume 0ms transmission delay for the ACK). **What is the throughput?**

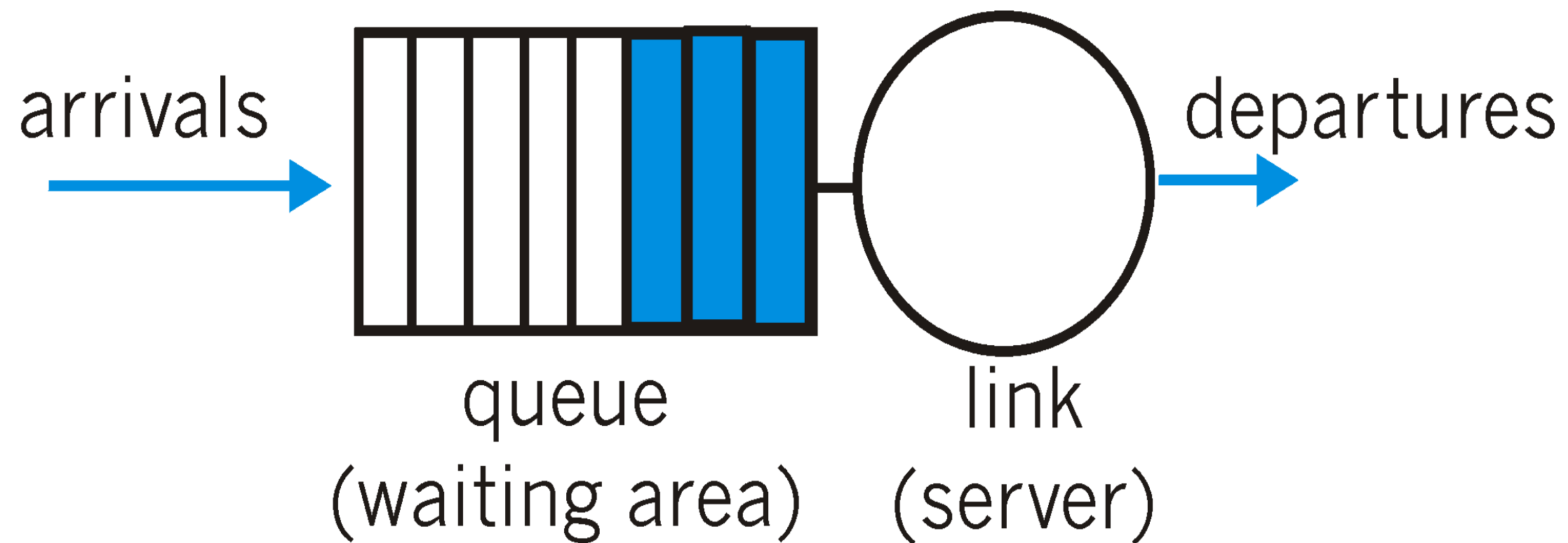
More Calculations – Throughput

- You are designing a satellite network. The satellites are 750km from the surface of the earth. The speed of light is 3×10^8 metres per second. Packets are 1250 Bytes. The network has a transfer rate (bandwidth) of 100Mbps. Assume that the **processing and queueing delays are 0**. Assume also that **you don't transmit the $(n+1)^{\text{th}}$ packet until you get a (very short) ACK for the n^{th} packet** (assume **0ms transmission delay for the ACK**). What is the throughput?
- **We transmit 1250 Bytes every RTT time**
- **Throughput = $(1250 * 8) / (5.1 * 10^{-3}) = 1\,960\,784.31 \text{ bps} = 1.96 \text{ Mbps}$**

Traffic Intensity

- How much data is arriving at the router?
- How much data can a router handle?
 - At what rate can the router forward data out?
- Queueing: when a router receives data faster than it can forward it

Traffic Intensity



Arrival Rate of 100 packets/second

Packets depart every 1 millisecond

Traffic Intensity Calculation

- Traffic intensity is determined by
 - Number of packets arriving per second (a)
 - Average packet size (L) in bits
 - Transmission rate: rate at which bits are disposed off per second (R)
- Traffic intensity: La/R
- Example:
 - Suppose a router is connected to a 1Mbps link. The router receives an average of 100 packets per second, averaging 500 bytes per packet. What is the traffic intensity?

Clicker Question

Suppose a router is connected to a 1Mbps link. The router receives an average of 100 packets per second, averaging 500 bytes per packet. What is the traffic intensity?

- A. 0.05
- B. 0.125
- C. 0.2
- D. 0.4
- E. 0.8

Clicker Answer

Suppose a router is connected to a 1 Mbps link. The router receives an average of 100 packets per second, averaging 500 bytes per packet. What is the traffic intensity?

$$\text{Traffic intensity} = (L \cdot a) / R = (500 \cdot 8 \cdot 100) / (1 \cdot 10^6) = 0.4$$

- A. 0.05
- B. 0.125
- C. 0.2
- D. 0.4**
- E. 0.8

Traffic Intensity Rationale

- Traffic intensity helps us understand how busy a link is
- Queueing delay is related to the intensity
 - Queueing delay is delay caused by waiting for the queue to clear
 - Packets arriving must wait for packets already there to leave

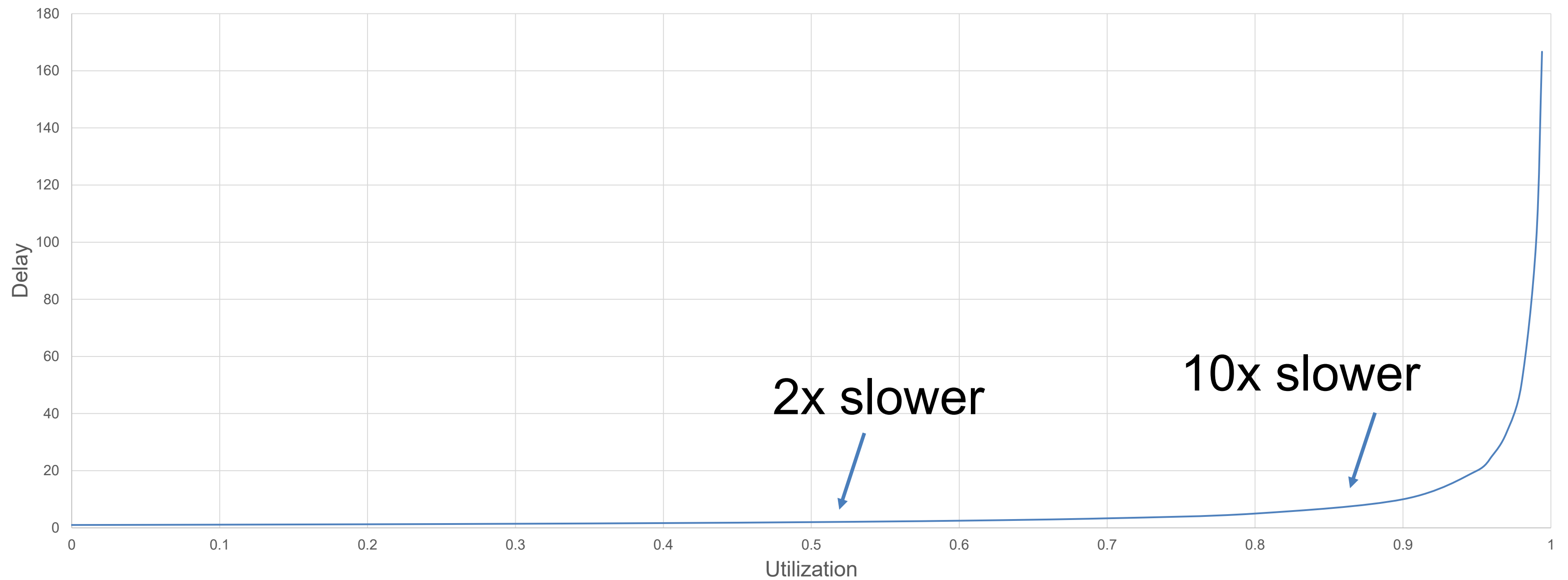
Traffic Intensity vs Queueing Delay

- Suppose λa bits/second arrive randomly for an outgoing link in a router
- Suppose that the router can transmit R bits/second
- Draw a graph of queueing delay vs traffic intensity
 - What does $\lambda a = R$ mean?
 - What does $\lambda a > R$ mean?
 - What does $\lambda a < R$ mean?
 - What does $\lambda a \ll R$ mean?

Queueing Problem

- **Packets are not spaced out evenly**
 - Spacing between packets is not deterministic
- Packets may not be served evenly
 - Link may be busy at times
 - Particularly for shared medium (e.g., radio signals)
- Higher intensity means higher probability that there is one or more packets in the queue

Delay vs Utilization



Traffic Intensity vs Queueing Delay

- Assuming packets arrive at an exponential distribution, delay is given by:

$$\text{Total Delay} = \frac{S}{1 - U}$$

- Where:
 - S is average service time when server is idle
 - U is server utilization (usually traffic intensity)
- Queueing delay is then:

$$\text{Queueing delay} = \frac{S}{1 - U} - S \text{ or } \frac{US}{1 - U}$$

Traffic Intensity vs Queueing Delay

- Assuming packets arrive at an exponential distribution, delay is given by:

$$\text{Total Delay} = \frac{S}{1 - U}$$

- Where:
 - S is average service time when server is idle
 - U is server utilization (usually traffic intensity)
- Queueing delay is then:

$$\text{Queueing delay} = \frac{S}{1 - U} - S \text{ or } \frac{US}{1 - U}$$

Some Observations

- Routers don't have infinite buffer space
- If packets arrive faster than they can be disposed off, they may have to be dropped
- Packets may also be corrupted in transit
 - These packets must be discarded, since their content is no longer valid
 - Even routing information (e.g., destination IP) may be corrupted

In-class Activity

- ICA22

Next Topic: Application Layer Protocols